

Лабораторная работа № 3. Анализ трафика в Wireshark

Данила Стариков
НПИбд-02-22

Российский университет дружбы народов имени Патриса Лумумбы

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Цель работы

- ▶ Изучение посредством Wireshark кадров Ethernet, анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.

MAC-адресация

```
➔ report git:(main) × ifconfig
enp1s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 80:e8:2c:1d:41:0a txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 3280394 bytes 201122127 (191.8 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3280394 bytes 201122127 (191.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.76 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::bdda:d902:27b9:c1b7 prefixlen 64 scopeid 0x20<link>
    ether a2:19:64:3c:02:6b txqueuelen 1000 (Ethernet)
    RX packets 5303260 bytes 7114543586 (6.6 GiB)
    RX errors 0 dropped 1243 overruns 0 frame 0
    TX packets 1414100 bytes 253962324 (242.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис.: Вывод команды ifconfig.

Анализ кадров канального уровня в Wireshark

```
sudo dnf install wireshark
sudo -i
groupadd wireshark
gpasswd -a dastarikov wireshark
chgrp wireshark /usr/bin/dumpcap
chmod 750 /usr/bin/dumpcap
setcap cap_net_raw,cap_net_admin=eip /usr/bin/dumpcap
```

Анализ кадров канального уровня в Wireshark

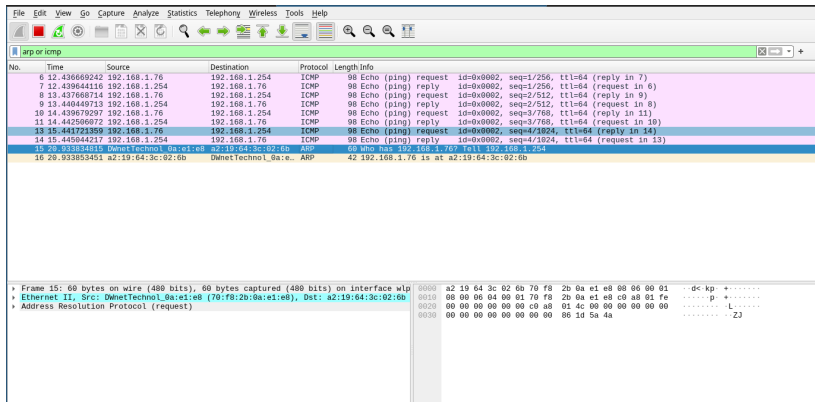


Рис.: Интерфейс Wireshark.

Анализ кадров канального уровня в Wireshark

```
→ report git:(main) × route -n
Kernel IP routing table
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface
0.0.0.0          192.168.1.254   0.0.0.0         UG    600    0      0 wlp2s0
192.168.1.0      0.0.0.0         255.255.255.0   U     600    0      0 wlp2s0
```

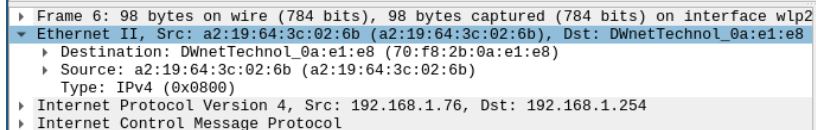
Рис.: Определение IP-адреса шлюза.

Анализ кадров канального уровня в Wireshark

```
→ report git:(main) × ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=64 time=3.02 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=64 time=2.86 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=64 time=2.89 ms
^C
--- 192.168.1.254 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 2.861/2.923/3.023/0.071 ms
→ report git:(main) ×
```

Рис.: Пингование шлюза.

Анализ кадров канального уровня в Wireshark



▶ Frame 6: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlp2
▼ Ethernet II, Src: a2:19:64:3c:02:6b (a2:19:64:3c:02:6b), Dst: DWnetTechnol_0a:e1:e8
 ▶ Destination: DWnetTechnol_0a:e1:e8 (70:f8:2b:0a:e1:e8)
 ▶ Source: a2:19:64:3c:02:6b (a2:19:64:3c:02:6b)
 Type: IPv4 (0x0800)
▶ Internet Protocol Version 4, Src: 192.168.1.76, Dst: 192.168.1.254
▶ Internet Control Message Protocol

Рис.: Информация об эхо-запросе.

Анализ кадров канального уровня в Wireshark

```
▶ Frame 7: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlp2
▼ Ethernet II, Src: DWnetTechnol_0a:e1:e8 (70:f8:2b:0a:e1:e8), Dst: a2:19:64:3c:02:6b
  ▼ Destination: a2:19:64:3c:02:6b (a2:19:64:3c:02:6b)
    Address: a2:19:64:3c:02:6b (a2:19:64:3c:02:6b)
      .... ..1. .... .. = LG bit: Locally administered address (this is N
      .... ..0 .... .. = IG bit: Individual address (unicast)
    ▶ Source: DWnetTechnol_0a:e1:e8 (70:f8:2b:0a:e1:e8)
      Type: IPv4 (0x0800)
    ▶ Internet Protocol Version 4, Src: 192.168.1.254, Dst: 192.168.1.76
    ▶ Internet Control Message Protocol
```

Рис.: Информация об эхо-ответе.

Анализ кадров канального уровня в Wireshark

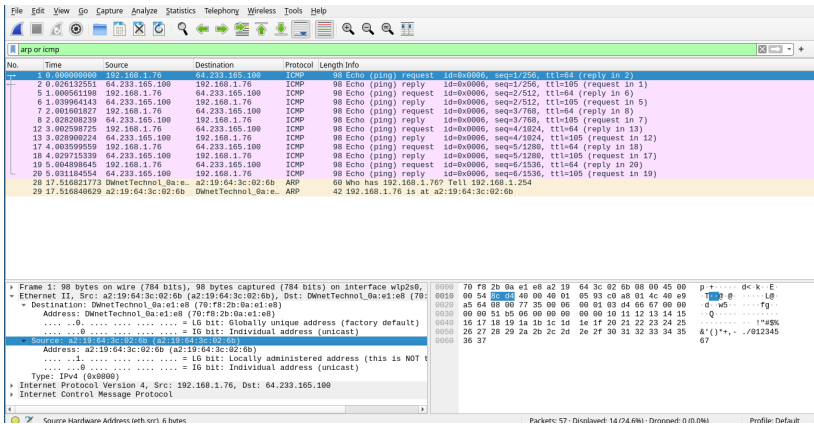


Рис.: Изучение трафика между устройством и google.com.

Анализ кадров канального уровня в Wireshark

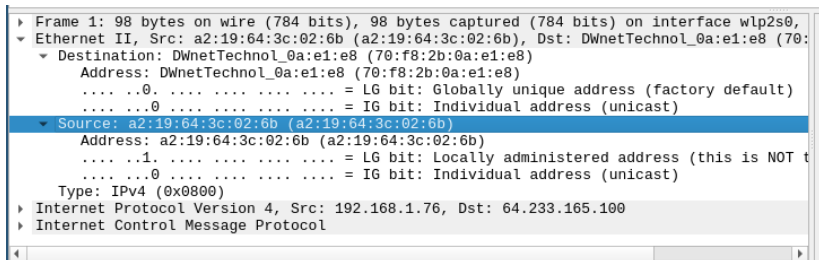


Рис.: Эхо-запрос к google.com

Анализ кадров канального уровня в Wireshark

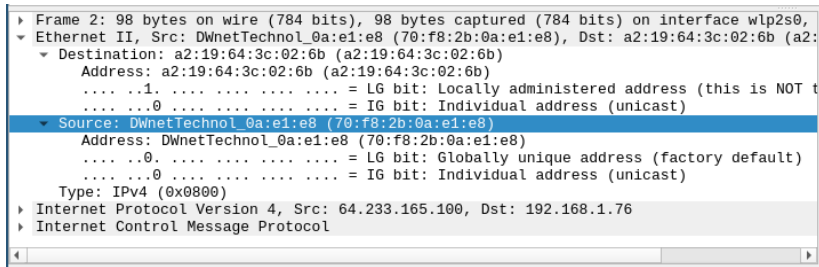


Рис.: Эхо-ответ от google.com

Анализ протоколов транспортного уровня в Wireshark

The image displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The main window is divided into three panes:

- Packet List:** Shows a list of captured packets. The first few packets are HTTP GET requests from 192.168.1.76 to 188.184.67.127. The status bar for the selected packet (Frame 19) is "200 OK".
- Packet Details:** Shows the hierarchical structure of the selected packet. The top level is "Ethernet II", followed by "Internet Protocol Version 4", "Transmission Control Protocol", and "Hypertext Transfer Protocol". The "Hypertext Transfer Protocol" section is expanded, showing the "GET / HTTP/1.1" request and the "200 OK" status.
- Packet Bytes:** Shows the raw data of the selected packet in hexadecimal and ASCII. The ASCII view shows the HTML content of the response, including the "first website" and "TheProject.html" links.

The filter bar at the top of the packet list shows the filter "http", indicating that only HTTP traffic is displayed. The status bar at the bottom of the interface shows "Packets: 618 / Displayed: 20 (3.2%)".

Рис.: Фильтрация трафика по протоколу HTTP.

Анализ протоколов транспортного уровня в Wireshark

The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, capture control, and analysis. The main window is divided into three panes:

- Packet List Pane:** Displays a list of captured packets. The first column shows the packet number, the second shows the time, the third shows the source and destination IP addresses, and the fourth shows the protocol and length. The list is filtered to show only DNS packets. The selected packet is packet 5, a DNS query from 192.168.1.76 to 192.168.1.254.
- Packet Details Pane:** Shows the hierarchical structure of the selected packet. It includes the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and Domain Name System (query) section. The packet is 83 bytes long and is a standard query for the domain 'info.cern.ch'.
- Packet Bytes Pane:** Displays the raw bytes of the packet in hexadecimal and ASCII. The packet is 83 bytes long and contains a standard query for the domain 'info.cern.ch'.

Рис.: Фильтрация трафика по протоколу UDP.

Анализ протоколов транспортного уровня в Wireshark

The image displays a Wireshark packet capture of a TCP handshake. The packet list on the left shows a sequence of packets from 188 to 199. Packet 188 is the SYN packet (Seq=572, Win=39424). Packet 189 is the SYN-ACK packet (Seq=616, Win=39424). Packet 190 is the ACK packet (Seq=572, Win=0). The packet details pane on the right shows the structure of the selected packet (188), including the Ethernet II header, Internet Protocol Version 4 header, and the TCP header. The TCP header fields are expanded, showing the SYN flag set, sequence number 572, window size 39424, and the SYN-ACK sequence number 616. The packet bytes pane on the right shows the raw data of the packet, including the Ethernet II header and the TCP header.

No.	Time	Source	Destination	Protocol	Length	Info
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	452	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	294	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	337	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	128	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TCP	66	443 -> 60888 [ACK] Seq=572 Ack=2065 Win=39424 Len=0 TSval=899922413 TSecr=2969537992
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	119	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	97	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TCP	66	443 -> 60888 [ACK] Seq=616 Ack=2096 Win=39424 Len=0 TSval=899922460 TSecr=2969537998
188.	0.23	192.168.1.76	178.248.233.33	TLSv1..	383	Application Data
188.	0.23	192.168.1.76	178.248.233.33	TCP	66	60888 -> 443 [ACK] Seq=2996 Ack=933 Win=64128 Len=0 TSval=2969538319 TSecr=899922683
188.	0.27	146.75.118.214	192.168.1.76	TCP	74	35148 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=854062248 TSecr=0 WS=128
188.	0.27	146.75.118.214	192.168.1.76	TCP	74	443 -> 35148 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1452 SACK_PERM TSval=465438761 TSecr=8
188.	0.27	146.75.118.214	192.168.1.76	TCP	66	35148 -> 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=854062288 TSecr=465438761
188.	0.27	146.75.118.214	192.168.1.76	TLSv1..	1164	Client Hello (SNI=www.twitch.tv)
188.	0.27	146.75.118.214	192.168.1.76	TCP	74	35160 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=854062314 TSecr=0 WS=128
188.	0.27	146.75.118.214	192.168.1.76	TCP	66	443 -> 35148 [ACK] Seq=1 Ack=1099 Win=146432 Len=0 TSval=465438804 TSecr=854062201
188.	0.27	146.75.118.214	192.168.1.76	TLSv1..	519	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data
188.	0.27	146.75.118.214	192.168.1.76	TCP	66	35148 -> 443 [ACK] Seq=1099 Ack=454 Win=64128 Len=0 TSval=854062330 TSecr=465438804
188.	0.27	146.75.118.214	192.168.1.76	TLSv1..	130	Change Cipher Spec, Application Data
188.	0.27	146.75.118.214	192.168.1.76	TLSv1..	236	Application Data
188.	0.27	146.75.118.214	192.168.1.76	TCP	74	443 -> 35160 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1452 SACK_PERM TSval=167438471 TSecr=8
188.	0.27	146.75.118.214	192.168.1.76	TCP	54	35160 -> 443 [RST] Seq=1 Win=0 Len=0

1010 = Header Length: 40 bytes (10)

▼ Flags: 0x002 (SYN)

- 000. = Reserved: Not set
- ...0 = Accurate ECN: Not set
-0 = Congestion Window Reduced: Not set
-0 = ECN-Echo: Not set
-0 = Urgent: Not set
-0 = Acknowledgment: Not set
-0 = Push: Not set
-0 = Reset: Not set
-1 = Syn: Set

▼ [Expert Info (Chat/Sequence): Connection establish request (SYN): server port 443]
[Severity level: Chat]
[Group: Sequence]

0000 70 f8 2b 0a e1 e8 a2 19 64 3c 02 6b 08 00 45 00 p+....d< k: E
0010 00 3c f2 48 40 00 40 06 70 5d c0 a8 01 4c 92 4b < H0 0. }....L K
0020 76 06 89 4c 01 bb 6a 8b 3b 81 00 00 00 00 a0 02 v...L.;.....K
0030 fa f0 29 55 00 00 02 04 05 04 04 02 00 0a 32 e7 ..JU....:..
0040 74 a8 00 00 00 00 01 03 03 07

Рис.: handshake протокола TCP.

Анализ протоколов транспортного уровня в Wireshark

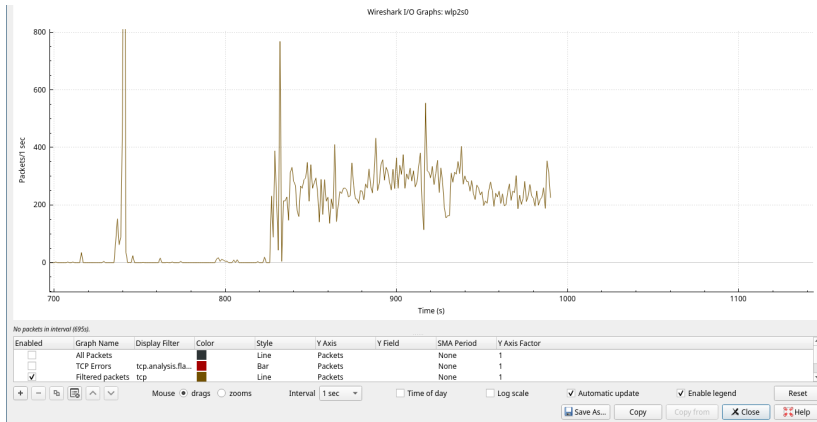


Рис.: График потока TCP пакетов.

Выводы

- ▶ В процессе выполнения лабораторной работы научились работать с программой Wireshark для анализа различных сетевых пакетов на транспортном и прикладном уровнях.